Math 110 — Week 1

Proofs, sets, fields, vector spaces, subspaces (Notes Ch. 1–2, LADR 1A–1C, HW1)

§1 This week

§1.1 How proofs are written (notes, opening section)

The question. Prove that for every integer m, if m^2 is even then m is even. Writing $m^2 = 2k$ gives no way to get at m itself. Which equivalent statement can be proved by a direct calculation instead?

A proof with every reason shown. The rational numbers are $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Claim (cancellation): if a, b, c all lie in $\mathbb{Q}$, or all in $\mathbb{R}$, and $a + b = a + c$, then $b = c$. Each line follows from the line before it by the stated rule:

$$\begin{aligned} b &= 0 + b && \text{additive identity} \\ &= ((-a) + a) + b && \text{additive inverse} \\ &= (-a) + (a + b) && \text{associativity} \\ &= (-a) + (a + c) && \text{hypothesis } a + b = a + c \\ &= ((-a) + a) + c && \text{associativity} \\ &= 0 + c = c && \text{additive inverse, additive identity.} \end{aligned}$$

Written for a reader, the same proof is one line: add $-a$ to both sides. Every later "add $-x$ to both sides" stands for this chain.

Example: direct proof, the product of two odd integers is odd. Unfold the definition, compute, and recognise the definition again.

1. "Odd" means of the form $2k + 1$ with k an integer. Write $m = 2a + 1$, $n = 2b + 1$.
2. $mn = (2a + 1)(2b + 1) = 2(2ab + a + b) + 1$.
3. $2ab + a + b$ is an integer, so mn has the form $2k + 1$: it is odd.

The definition is used twice: once to turn the hypothesis into something to calculate with, once to know what the answer has to look like. With $m = n$: m odd $\Rightarrow m^2$ odd.

Contrapositive. $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ are true or false together. The contrapositive of "m odd $\Rightarrow m^2$ odd" is "m^2 not odd $\Rightarrow m$ not odd", that is, m^2 even $\Rightarrow m$ even.

The question answered. Prove the contrapositive, "m odd $\Rightarrow m^2$ odd": it is the case $m = n$ of the direct proof above.

If and only if means two implications, and each may need its own argument. Example: m even $\iff m^2$ even. ($\Leftarrow$) is the contrapositive above. ($\Rightarrow$): $m = 2k$ gives $m^2 = 2(2k^2)$.

Disproving a statement. A statement about all objects of some kind is false as soon as one object satisfies the hypothesis and fails the conclusion. Example: "if $x^2 = y^2$ for real x, y, then $x = y$" is false, because $x = 1$, $y = -1$ give $x^2 = y^2 = 1$ and $x \neq y$.

Proposition (Prop. 1)
$\sqrt{2}$ is irrational.

The idea

"Irrational" is a negative statement (there is *no* fraction equal to $\sqrt{2}$), so there is nothing to compute with directly. Assuming the opposite gives an equation: $\sqrt{2} = m/n$, so $m^2 = 2n^2$. A fraction can be written in lowest terms, so m, n are not both even. The equation $m^2 = 2n^2$ talks about evenness, and "m^2 even $\Rightarrow$ m even" is available. So follow evenness through the equation and aim to show both m and n are even.

Proof. 1. Suppose $\sqrt{2} = m/n$ with integers m, n, $n \neq 0$, in lowest terms. So at most one of m, n is even.

2. Squaring: $m^2 = 2n^2$. So m^2 is even, hence m is even. Write $m = 2r$.

3. Then $4r^2 = 2n^2$, so $n^2 = 2r^2$ is even, hence n is even.

4. Both m and n are even, contradicting step 1.

□

Equality of sets: two inclusions. $A = B$ means the same elements, so prove $A \subseteq B$ and $B \subseteq A$, each by taking an arbitrary element.

Example: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$**.**

1. ($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$, and $x \in B$ or $x \in C$. In the first case $x \in A \cap B$, in the second $x \in A \cap C$.

2. ($\supseteq$) Let $x \in (A \cap B) \cup (A \cap C)$. In either case $x \in A$, and $x \in B$ or $x \in C$, so $x \in A \cap (B \cup C)$.

Proposition (Prop. 2: induction)
Let $P(n)$ be a statement for each integer $n \geq 1$. If $P(1)$ holds, and $P(n) \Rightarrow P(n+1)$ for every $n \geq 1$, then $P(n)$ holds for all $n \geq 1$.

The idea

The chain $P(1)$, $P(1) \Rightarrow P(2)$, $P(2) \Rightarrow P(3)$, ... reaches every n, but it is an infinite list of steps, and a proof is a finite list. So argue about a failure instead: if some $P(n)$ were false, look at the smallest such n. The statement just before it is true, and the hypothesis carries truth one step forward, so a smallest failure cannot exist. This uses the **well-ordering property**: every nonempty set of positive integers has a smallest element.

Proof. 1. Suppose $P(n)$ is false for at least one n. By well-ordering there is a smallest such n; call it m.

2. $P(1)$ is true by hypothesis, so $m \neq 1$, and $m - 1$ is a positive integer.

3. $m - 1 < m$, so $P(m-1)$ is true, because m is the smallest failure.

4. Applying "$P(n) \Rightarrow P(n+1)$" with $n = m - 1$ shows that $P(m)$ is true. This contradicts the choice of m, so no $P(n)$ is false.

□

Example: $1 + 3 + \cdots + (2n-1) = n^2$**.** Let $P(n)$ be the statement $\sum_{i=1}^{n}(2i-1) = n^2$. $P(1)$ says $1 = 1^2$. Assume $P(n)$. Then

$$\sum_{i=1}^{n+1}(2i-1) = \sum_{i=1}^{n}(2i-1) + (2n+1) = n^2 + 2n + 1 = (n+1)^2,$$

which is $P(n+1)$. The hypothesis $P(n)$ is used exactly once, to replace the sum up to $2n-1$ by n^2.

Use induction when the case $n+1$ can be reduced to the case n. In this course that is usually induction on a dimension or on the length of a list.

Shortcuts

- **Contrapositive**: to prove $P \Rightarrow Q$, prove $\neg Q \Rightarrow \neg P$. Reason: the two have the same truth value. Condition: use it when Q is a negative statement or P gives nothing to compute with. Instance: "m^2 even $\Rightarrow m$ even" from "m odd $\Rightarrow m^2$ odd".
- **Disproof by one example**: a statement "for all x, ..." is false as soon as one x meets the hypothesis and fails the conclusion. Instance: "$x^2 = y^2 \Rightarrow x = y$" fails for $x = 1$, $y = -1$.
- **Set equality**: prove the two inclusions separately, each by taking an arbitrary element. Instance: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ above.

Traps

- Proving the converse instead of the claim: "$x = 2 \Rightarrow x^2 = 4$" is true, but its converse "$x^2 = 4 \Rightarrow x = 2$" is false ($x = -2$).
- An induction step must work for every $n \geq 1$. Claim: in any finite group of cows, all cows have the same colour. The "proof": one cow has one colour; for $n+1$ cows, remove Bessie, and the other n have one colour; put her back and remove Angus, and again the other n have one colour; Bessie shares a colour with the cows that stayed both times, so all agree. The step from $n = 1$ to $n = 2$ fails: with two cows, no cow stays in the field both times, so nothing links the two colours.

§1.2 Sets and functions (Ch. 1)

The question. Let $f\colon \{1,2,3\} \to \{1,2,3\}$ with $f(1) = 2$, $f(2) = 3$, $f(3) = 2$. Is f injective? Is it surjective? Can a map between two sets with 3 elements be injective without being surjective?

Definition (Def. 1, 2)**.** A **set** is a collection of distinguishable objects, its **elements**. $x \in M$: x is an element of M. $N \subseteq M$: every element of N is in M. A **function** $f\colon M \to N$ assigns an element $f(x) \in N$ to every $x \in M$. In the notes $N \subset M$ also means $N \subseteq M$ (some books use $\subset$ only for a proper subset). A set is written by listing its elements, $\{2,4,6\}$, or by a property after a colon or a bar: $\{x : x \text{ even}\} = \{x \mid x \text{ even}\}$.

Example 1. $\{x : x \in \{1,\dots,6\},\ x \text{ even}\} = \{2,4,6\}$.

- For subsets M, N of A: complement $M^c = \{x \in A : x \notin M\}$; union $M \cup N$ ($x \in M$ or $x \in N$); intersection $M \cap N$ ($x \in M$ and $x \in N$); difference $M \setminus N = M \cap N^c$; product $M \times N = \{(x,y) : x \in M, y \in N\}$. Pairs are ordered: $(1,2) \neq (2,1)$.
- f is **injective** (one-to-one) if $f(x) = f(y) \Rightarrow x = y$; **surjective** (onto) if every $z \in N$ is $f(x)$ for at least one x; **bijective** if both.
- For $U \subseteq M$ and $V \subseteq N$: the **image** $f(U) = \{f(x) : x \in U\}$ and the **preimage** $f^{-1}(V) = \{x \in M : f(x) \in V\}$.

Proposition (Ex. 2: pigeonhole principle)

If A, B are finite sets with the same number of elements and $f\colon A \to B$, then f is injective $\iff$ f is surjective.

The idea

Count the values f takes. Say A and B have n elements. f produces n values $f(a)$, one for each a. If two of them coincide, at most $n-1$ different elements of B are reached, so one is missed. If none coincide, n different elements of B are reached, which is all of B.

Proof. 1. Let n be the number of elements of A and of B, and let k be the number of different values among the $f(a)$. So $k \leq n$.

2. f is injective exactly when no two values coincide, that is, $k = n$.

3. f is surjective exactly when all n elements of B are values, that is, $k = n$.

□

The question answered. $f(1) = f(3)$, so f is not injective, and 1 is not a value of f, so f is not surjective. By the pigeonhole principle, a map between two 3-element sets is injective exactly when it is surjective, so the answer to the last question is no.

Shortcuts

- **Finite sets of equal size**: check only one of injective and surjective. Reason: the pigeonhole principle. Condition: A and B finite with the same number of elements. Instance: $f(x) = 4 - x$ on $\{1, 2, 3\}$ is injective ($4 - x = 4 - y$ gives $x = y$), so it is surjective.
- **Images and preimages**: to prove an inclusion, take one element and unfold $f(U) = \{f(x) : x \in U\}$ and $f^{-1}(V) = \{x : f(x) \in V\}$. Instance: $x \in U$ gives $f(x) \in f(U)$, so $x \in f^{-1}(f(U))$.

Traps

- For infinite sets, injective does not give surjective: $n \mapsto n + 1$ on $\{1, 2, 3, \dots\}$ is injective and misses 1.
- $f(f^{-1}(V))$ can be smaller than V, and $f^{-1}(f(U))$ larger than U: for $f(x) = x^2$ on $\mathbb{R}$, $f(f^{-1}(\{-4, 4\})) = f(\{-2, 2\}) = \{4\}$ and $f^{-1}(f(\{2\})) = \{-2, 2\}$.

§1.3 Fields (Ch. 1)

The question. Which number systems are allowed as scalars? We need to add, subtract, multiply, and divide by anything nonzero. Do $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}$ and the two-element system $\{0, 1\}$ with $1 + 1 = 0$ allow this?

Definition (Def. 3). A **field** is a set $\mathbb{F}$ with two different elements $0, 1$ and operations $+$, $\cdot$ such that for all $x, y, z \in \mathbb{F}$:

(i) $x + y = y + x$ and $xy = yx$;
(ii) $x + (y + z) = (x + y) + z$ and $x(yz) = (xy)z$;
(iii) $x + 0 = x$;
(iv) $1 \cdot x = x$;
(v) $x(y + z) = xy + xz$;
(vi) every x has an additive inverse $-x$ with $(-x) + x = 0$;
(vii) every $x \neq 0$ has a multiplicative inverse x^{-1} with $x^{-1}x = 1$.

Example 3: $\mathbb{Q}$ and $\mathbb{R}$ are fields, $\mathbb{Z}$ is not. $\mathbb{Q}$ and $\mathbb{R}$ satisfy all seven axioms. $\mathbb{Z}$ satisfies (i)–(vi), but only ± 1 have multiplicative inverses in $\mathbb{Z}$: $2x = 1$ has no integer solution.

Proposition (Uniqueness of identities and inverses)

In a field, the additive identity, the multiplicative identity, additive inverses and multiplicative inverses are unique.

The idea

To show there is only one object with a property, take two objects with it and show they are equal. For identities: combine the two candidates, $0+0'$. Read with $0'$ as the identity, this is 0; read with 0 as the identity, this is $0'$. For inverses: put x between its two inverses, $(-x)+x+(-x)'$, and group it in two ways.

Proof. 1. **The additive identity is unique.** Let $0'$ also satisfy $x+0'=x$ for all x. By the property of $0'$ (with $x=0$), axiom (i), and axiom (iii) (with $x=0'$),

$$0 = 0+0' = 0'+0 = 0'.$$

2. **The multiplicative identity is unique.** Let $1'$ also satisfy $1'\cdot x = x$ for all x. By the property of $1'$ (with $x=1$), axiom (i), and axiom (iv) (with $x=1'$),

$$1 = 1'\cdot 1 = 1\cdot 1' = 1'.$$

3. **Additive inverses are unique.** If $-x$ and $(-x)'$ are both additive inverses of x, then by axioms (iii), (vi) with (i), (ii), (vi) and (iii) with (i):

$$-x = -x+0 = -x+(x+(-x)') = (-x+x)+(-x)' = 0+(-x)' = (-x)'.$$

4. **Multiplicative inverses are unique.** If $x\neq 0$ and y, y' both satisfy $yx = 1 = y'x$, then by axioms (iv) with (i), (vii) with (i), (ii), (vii) and (iv):

$$y = y\cdot 1 = y(xy') = (yx)y' = 1\cdot y' = y'.$$

□

Proposition (HW1 P3: no zero divisors)

In a field, $0\cdot a = 0$ for all a, and if $ab=0$ then $a=0$ or $b=0$.

Proof. 1. $0\cdot a = (0+0)a = 0\cdot a + 0\cdot a$. Add $-(0\cdot a)$ to both sides (cancellation): $0 = 0\cdot a$.

2. Suppose $ab=0$ and $a\neq 0$. Multiply by a^{-1}:

$$b = 1\cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1}\cdot 0 = 0.$$

□

The second step uses axiom (vii). Without inverses this can fail: in the arithmetic of remainders on division by 6 (not a field, since 2 has no inverse), $2\cdot 3 = 6$ has remainder 0.

Complex numbers (Ex. 4). $\mathbb{C} = \{a+bi : a,b\in\mathbb{R}\}$ with $i^2=-1$; add and multiply by the usual rules. $\mathbb{R}\subseteq\mathbb{C}$ via $x = x+0i$. In $\mathbb{C}$ the equation $x^2+1=0$, which has no real solution, has the two solutions $\pm i$.

Example 4: arithmetic in $\mathbb{C}$. Expand and use $i^2=-1$:

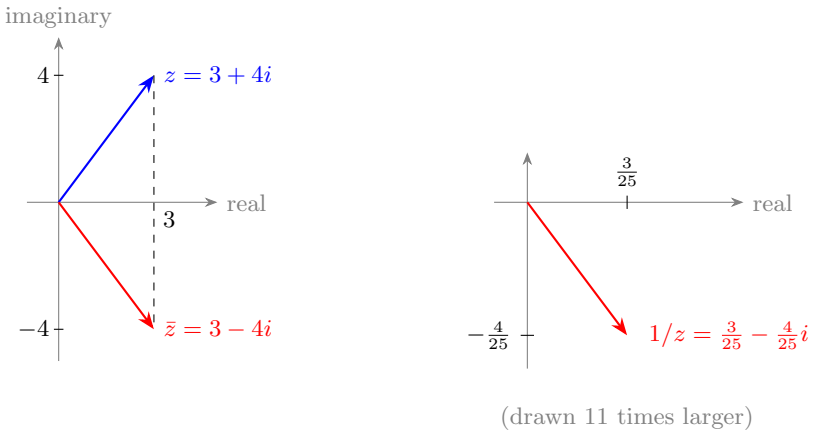
$$(1+i)^3 = 1+3i+3i^2+i^3 = -2+2i, \qquad (5+i)(4+3i) = 20+19i+3i^2 = 17+19i.$$

- **Conjugate**: $\overline{a+bi} = a-bi$. Then $(a+bi)(a-bi) = a^2+b^2$. The **absolute value** is $|z| = \sqrt{a^2+b^2}$ for $z=a+bi$, so $z\bar{z} = |z|^2$.

- $\mathbb{C}$ **is a field.** The other axioms follow from the rules of arithmetic; inverses need a formula. If $a+bi \neq 0$ then $a^2+b^2 > 0$, and

$$(a+bi)\cdot\frac{a-bi}{a^2+b^2} = \frac{a^2+b^2}{a^2+b^2} = 1.$$

For example, $1/(3+4i) = (3-4i)/25$. In the plane, $\bar{z}$ is the reflection of z across the real axis, and $1/z = \bar{z}/|z|^2$ points the same way as $\bar{z}$ with length $1/|z|$.



Left: conjugation reflects $3+4i$ to $3-4i$. Right: $1/z$ lies on the ray through $\bar{z}$ at distance $1/|z| = 1/5$ from 0, because $1/z = \bar{z}/(z\bar{z}) = \bar{z}/25$.

Why $\mathbb{C}$ matters. The cubic $x^3 - 15x - 4 = 0$ has the real root 4, but the cubic formula gives $\sqrt[3]{2+11i} + \sqrt[3]{2-11i}$. Each nonzero complex number has three cube roots. Since $(2\pm i)^3 = 2 \pm 11i$, choosing the cube roots $2+i$ and $2-i$ gives $(2+i)+(2-i) = 4$: complex numbers appear inside a calculation whose answer is real. Formulas by radicals exist for equations of degree 2, 3 and 4, but not for every equation of degree 5; that is a statement about formulas, not about whether roots exist. Every nonconstant polynomial with complex coefficients has a complex root (the fundamental theorem of algebra, cited here and proved later in the course), while x^2+1 has no real root.

Proposition (Prop. 3: conjugation)
For $w, z \in \mathbb{C}$: (i) $\overline{w+z} = \bar{w}+\bar{z}$; (ii) $\overline{wz} = \bar{w}\,\bar{z}$; (iii) $\bar{z} = z \iff z$ is real; (iv) $z\bar{z}$ is real and ≥ 0.

Proof. Write $w = a+bi$, $z = c+di$.

1. The sum is $w+z = (a+c)+(b+d)i$, so
$$\overline{w+z} = (a+c)-(b+d)i = (a-bi)+(c-di) = \bar{w}+\bar{z}.$$
2. $wz = (ac-bd)+(ad+bc)i$, so $\overline{wz} = (ac-bd)-(ad+bc)i$, and this equals
$$\bar{w}\bar{z} = (a-bi)(c-di) = (ac-bd)-(ad+bc)i.$$
3. $z-\bar{z} = 2di$, which is 0 exactly when $d = 0$.
4. $z\bar{z} = c^2+d^2 \geq 0$.

□

The field $\mathbb{F}_2$. The set $\{0,1\}$ with the tables below is a field; it is the arithmetic of even (0) and odd (1) numbers, so $1+1=0$ says that odd plus odd is even.

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

In it $2x = x+x = (1+1)x = 0\cdot x = 0$ for every x, so $(x+1)^2 = x^2+2x+1 = x^2+1$ (HW1 P7). Most theorems in this course hold over any field; a result that needs $\mathbb{R}$ or $\mathbb{C}$ says so.

The question answered. $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{F}_2$ satisfy every axiom of Def. 3 (for $\mathbb{C}$ the inverse is given by the conjugate formula above, for $\mathbb{F}_2$ by the tables). $\mathbb{Z}$ fails axiom (vii).

Shortcuts

- **Dividing by a complex number**: multiply top and bottom by the conjugate of the bottom. Reason: $z\bar{z} = |z|^2$ is real (Prop. 3(iv)). Condition: $z \neq 0$. Instance:

$$\frac{2+i}{1-i} = \frac{(2+i)(1+i)}{(1-i)(1+i)} = \frac{1+3i}{2}.$$

- **Showing $\omega^3 = 1$ without cubing**: $\omega = (-1+\sqrt{3}\,i)/2$ is a root of x^2+x+1 (quadratic formula), and $x^3-1 = (x-1)(x^2+x+1)$. Reason: a root of one factor is a root of the product. Instance: $\omega^3 - 1 = (\omega-1)\cdot 0 = 0$.
- **Squaring over $\mathbb{F}_2$**: $(a+b)^2 = a^2+b^2$. Reason: the cross term is $2ab = (1+1)ab = 0$. Instance: $(x+1)^4 = (x^2+1)^2 = x^4+1$.
- **Uniqueness proofs**: take two objects with the property and combine them. Reason: if both have the property, the combination equals each of them. Instance: two additive identities satisfy $0 = 0+0' = 0'$.

Traps

- Without multiplicative inverses, $ab = 0$ does not force $a = 0$ or $b = 0$: for remainders on division by 6, $2\cdot 3 = 6$ has remainder 0.
- $\mathbb{Z}$ is not a field: $2x = 1$ has no solution in $\mathbb{Z}$.
- "The" cube root of a complex number needs a choice: $(2+i)^3 = 2+11i$, and $\big((2+i)\omega\big)^3 = (2+i)^3\omega^3 = 2+11i$ as well, with $(2+i)\omega \neq 2+i$.

§1.4 Vector spaces (Ch. 2)

The question. On $\mathbb{R}^2$ keep the usual addition, but define scalar multiplication by $\lambda\cdot(x,y) = (\lambda x, 0)$. Which axioms of Def. 4 below hold? Is this a vector space?

Definition (Def. 4 / LADR 1.20)**.** An **$\mathbb{F}$-vector space** is a set V with addition $V\times V\to V$ and scalar multiplication $\mathbb{F}\times V\to V$ such that for all $u,v,w\in V$ and $\lambda,\mu\in\mathbb{F}$:

(i) $v+w = w+v$;

(ii) $(v+w)+u = v+(w+u)$;

(iii) there is $0\in V$ with $v+0 = v$ for all v;

(iv) every v has $-v\in V$ with $v+(-v) = 0$;

(v) $(\lambda\mu)v = \lambda(\mu v)$;

(vi) $\lambda(v+w) = \lambda v+\lambda w$ and $(\lambda+\mu)v = \lambda v+\mu v$;

(vii) $1\cdot v = v$.

- The zero vector is unique: if $0'$ also satisfies $v+0' = v$ for all v, then $0 = 0+0' = 0'+0 = 0'$ (property of $0'$, axiom (i), axiom (iii)). Additive inverses are unique: if $v+w = 0$ and $v+w' = 0$, then

$$w = w+0 = w+(v+w') = (w+v)+w' = 0+w' = w'$$

  by axioms (iii), (ii) with the hypothesis, (ii), (i) with the hypothesis, and (iii) with (i) (LADR 1.26, 1.27).
- Axiom (vii) is needed: on $\mathbb{R}^2$ with the usual addition and $\lambda\cdot v = 0$ for all λ, v, axioms (i)–(vi) hold.

Example 5: $\mathbb{F}^n$. Lists $(x_1, \dots, x_n)$ with componentwise operations. Zero vector $(0, \dots, 0)$; do not confuse it with the number 0. $\mathbb{F} = \mathbb{F}^1$.

Example 6: $\mathbb{F}^S$. All functions $f\colon S \to \mathbb{F}$, with $(f+g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$. Zero vector: the constant function 0. For $S = \{1, \dots, n\}$ this is $\mathbb{F}^n$; for $S = \{1, 2, 3, \dots\}$ it is the space of sequences $\mathbb{F}^\infty$.

Example 7: $\mathbb{C}$ over $\mathbb{R}$, and $\mathbb{C}$, $\mathbb{R}$ over $\mathbb{Q}$. $\mathbb{C}$ is a real vector space (allow only real scalars). $\mathbb{C}$ and $\mathbb{R}$ are $\mathbb{Q}$-vector spaces.

The question answered. Axioms (i)–(iv) involve only addition, which is the usual one. Axioms (v) and (vi) hold:

$$\begin{aligned}(\lambda\mu)\cdot(x,y) &= (\lambda\mu x, 0) = \lambda\cdot(\mu x, 0) = \lambda\cdot\big(\mu\cdot(x,y)\big),\\ \lambda\cdot\big((x,y)+(x',y')\big) &= (\lambda x + \lambda x', 0) = \lambda\cdot(x,y) + \lambda\cdot(x',y'),\\ (\lambda+\mu)\cdot(x,y) &= (\lambda x + \mu x, 0) = \lambda\cdot(x,y) + \mu\cdot(x,y).\end{aligned}$$

Axiom (vii) fails: $1\cdot(0,1) = (0,0) \neq (0,1)$. So this is not a vector space, and (vii) is the only axiom that detects it.

Proposition (Prop. 4 / LADR 1.30–1.32)
(i) $0\cdot v = 0$ for every $v \in V$. (ii) $\lambda\cdot 0 = 0$ for every $\lambda \in \mathbb{F}$. (iii) $(-1)v = -v$.

The idea

These are not axioms, so they need proofs. The only axiom that connects the number 0 (or addition of numbers) with vectors is the distributive law (vi). So write 0 as $0+0$ and distribute: this produces an equation of the form $x = x + x$, and subtracting x leaves $x = 0$. For (iii), $-v$ is *defined* as the vector that gives 0 when added to v, so check that $(-1)v$ does that.

Proof. 1. $0\cdot v = (0+0)v = 0\cdot v + 0\cdot v$. Add $-(0\cdot v)$ to both sides (cancellation, which works in V by the same chain as above with axioms (i)–(iv)): $0 = 0\cdot v$.

2. Since $0 = 0+0$, the distributive axiom gives

$$\lambda\cdot 0 = \lambda(0+0) = \lambda\cdot 0 + \lambda\cdot 0.$$

Adding $-(\lambda\cdot 0)$ to both sides gives $0 = \lambda\cdot 0$.

3. Using $1\cdot v = v$, the distributive axiom, and (i):

$$(-1)v + v = (-1)v + 1\cdot v = (-1+1)v = 0\cdot v = 0.$$

So $(-1)v$ is an additive inverse of v. Additive inverses are unique, so $(-1)v = -v$.

□

Proposition (HW1 P4 / LADR 1B.2)
If $av = 0$ with $a \in \mathbb{F}$, $v \in V$, then $a = 0$ or $v = 0$.

Proof. 1. If $a = 0$ there is nothing to prove. Assume $a \neq 0$; then a^{-1} exists (field axiom (vii)).

2. By vector-space axioms (vii) and (v), the hypothesis $av = 0$, and Prop. 4(ii),

$$v = 1\cdot v = (a^{-1}a)v = a^{-1}(av) = a^{-1}\cdot 0 = 0.$$

□

Shortcuts

- **Function spaces satisfy the axioms pointwise**: each axiom in $\mathbb{F}^S$ is an equality of functions, which holds at every $x \in S$ because the same rule holds in $\mathbb{F}$. Condition: the operations are defined pointwise. Instance: axiom (i) holds because, for every x,

$$(f+g)(x) = f(x) + g(x) = g(x) + f(x) = (g+f)(x).$$

- **Cancelling a nonzero scalar**: $av = aw$ with $a \neq 0$ gives $v = w$. Reason: $a(v-w) = av - aw = 0$ (axiom (vi) and Prop. 4(iii)), then HW1 P4. Instance: $3v = 3w$ in a real vector space gives $v = w$.

Traps

- "$(-1)v = -v$ holds by definition" is wrong: $-v$ is defined by $v + (-v) = 0$, and $(-1)v = -v$ needs axiom (vii). On $\mathbb{R}^2$ with the usual addition and $\lambda \cdot v = 0$ for all λ, v, axioms (i)–(vi) hold, but $(-1)\cdot(1,0) = (0,0) \neq (-1,0)$.
- Changing an operation can destroy a vector space: $\mathbb{R}^2$ with $\lambda \cdot (x,y) = (\lambda x, 0)$ fails axiom (vii) (the question of this section).

§1.5 Subspaces (Ch. 2)

The question. In $\mathbb{R}^2$, which of these are subspaces: the line $y = 2x$; the line $y = 2x + 1$; the union of the two axes; the quadrant $\{(x,y) : x \geq 0,\ y \geq 0\}$?

Definition (Def. 5 / LADR 1.34)**.** A subset $U \subseteq V$ is a **subspace** if (i) $0 \in U$, and (ii) $\lambda v + w \in U$ for all $v, w \in U$, $\lambda \in \mathbb{F}$.

- Condition (ii) is "closed under addition" ($\lambda = 1$) and "closed under scalar multiplication" ($w = 0$) together. A subspace is a vector space with the operations of V: by (ii) with $\lambda = 1$ and with $w = 0$, sums and scalar multiples of vectors of U lie in U; $0 \in U$ by (i); $-v = (-1)v \in U$ (Prop. 4(iii)); and every axiom, being an equation that holds for all vectors of V, holds in particular for the vectors of U.
- **To show a set is a subspace**: check 0, then take v, w in the set and check $\lambda v + w$. **To show it is not**: exhibit $0 \notin U$, or two specific vectors whose sum leaves U, or a specific multiple that leaves U.

Example 8: continuous and differentiable functions. In $\mathbb{R}^{(0,1)}$, the continuous functions form a subspace, and so do the differentiable functions (the zero function is continuous and differentiable, and sums and scalar multiples of continuous, resp. differentiable, functions are continuous, resp. differentiable).

Example 9: a recurrence subspace. The real sequences $(a_0, a_1, a_2, \dots)$ (functions $\{0,1,2,\dots\} \to \mathbb{R}$, Ex. 6) with $a_{n+2} = a_{n+1} + a_n$ for all n form a subspace. The zero sequence satisfies the recurrence. If a and b satisfy it, so does $\lambda a + b$, since adding λ times the recurrence for a to the recurrence for b gives

$$\lambda a_{n+2} + b_{n+2} = (\lambda a_{n+1} + b_{n+1}) + (\lambda a_n + b_n).$$

The Fibonacci sequence is one element. Each such sequence is determined by a_0, a_1 (by induction on n: a_0, a_1 are given, and a_{n+2} is computed from a_{n+1} and a_n).

Examples (LADR 1.35): subspaces given by conditions.

- $\{(x_1,x_2,x_3,x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$ is a subspace exactly when $b = 0$. If $b \neq 0$, then 0 fails $0 = 5 \cdot 0 + b$. If $b = 0$, $v_3 = 5v_4$ and $w_3 = 5w_4$, then $\lambda v_3 + w_3 = 5(\lambda v_4 + w_4)$.
- The differentiable $f : \mathbb{R} \to \mathbb{R}$ with $f'(2) = b$ form a subspace of $\mathbb{R}^{\mathbb{R}}$ exactly when $b = 0$, by the same two checks and $(\lambda f + g)'(2) = \lambda f'(2) + g'(2)$.

- The sequences $(a_1, a_2, \dots)$ of complex numbers with $\lim a_n = 0$ form a subspace of $\mathbb{C}^\infty$, because $\lim(\lambda a_n + b_n) = \lambda \lim a_n + \lim b_n$.

The question answered. The line $y = 2x$ is a subspace: it contains 0, and if $v_2 = 2v_1$ and $w_2 = 2w_1$ then $\lambda v_2 + w_2 = 2(\lambda v_1 + w_1)$. The line $y = 2x + 1$ is not: it misses $(0, 0)$. The union of the axes is not: $(1, 0) + (0, 1) = (1, 1)$ is on neither axis. The quadrant is not: it is closed under addition, but $(-1) \cdot (1, 0) = (-1, 0)$ leaves it.

Proposition (Prop. 5 / LADR 1C.10)

If U, W are subspaces of V, so is $U \cap W$.

Proof. 1. $0 \in U \cap W$, because $0 \in U$ and $0 \in W$.

2. Let $v, w \in U \cap W$ and $\lambda \in \mathbb{F}$. Since U is a subspace containing v, w, $\lambda v + w \in U$. Since W is a subspace containing v, w, $\lambda v + w \in W$. So $\lambda v + w \in U \cap W$.

□

Proposition (Prop. 6 / LADR 1.40)

If U, W are subspaces of V, then $U + W = \{u + w : u \in U, w \in W\}$ is a subspace, and it is the smallest subspace containing $U \cup W$.

The idea

The union $U \cup W$ is usually not a subspace: for the two axes in $\mathbb{R}^2$, $(1, 0) + (0, 1)$ is in neither. Any subspace containing U and W has to contain every $u + w$. So the set of all such sums is forced; the proposition says that this set is already closed, so nothing more has to be added.

Proof. 1. $0 = 0 + 0 \in U + W$.

2. For $u_1 + w_1$, $u_2 + w_2 \in U + W$ and $\lambda \in \mathbb{F}$, regroup:

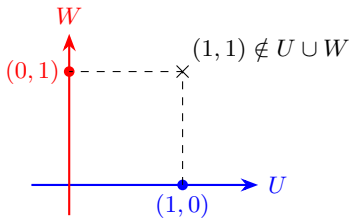
$$\lambda(u_1 + w_1) + (u_2 + w_2) = (\lambda u_1 + u_2) + (\lambda w_1 + w_2),$$

which lies in $U + W$ because $\lambda u_1 + u_2 \in U$ and $\lambda w_1 + w_2 \in W$.

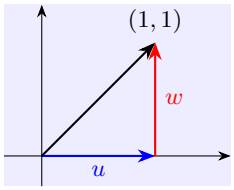
3. It contains $U \cup W$: $u = u + 0$ and $w = 0 + w$.

4. Smallest: if a subspace X contains $U \cup W$, then for $u \in U$, $w \in W$ both lie in X, so $u + w \in X$. Hence $U + W \subseteq X$.

□

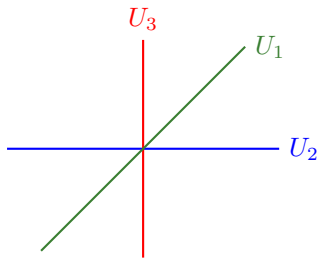


$U \cup W$ is the two axes.



$U + W = \mathbb{R}^2$: $(x, y) = (x, 0) + (0, y)$.

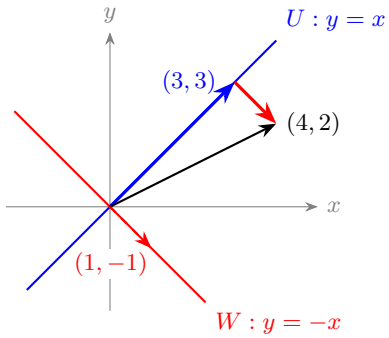
A subspace containing both axes must contain $(1, 1) = (1, 0) + (0, 1)$, which is in neither axis. In the same way it contains every $(x, y) = (x, 0) + (0, y)$, so here $U + W$ is all of $\mathbb{R}^2$.



Three different lines through 0 in $\mathbb{R}^2$. $U_2 + U_3 = \mathbb{R}^2$, so $U_1 \cap (U_2 + U_3) = U_1$. But $U_1 \cap U_2 = U_1 \cap U_3 = \{0\}$, so $(U_1 \cap U_2) + (U_1 \cap U_3) = \{0\}$. The distributive law for sets fails for subspaces when union is replaced by sum.

Definition (Def. 6)**.** The sum of U and W is **direct**, written $U \oplus W$, if $U \cap W = \{0\}$. A sum $U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_k \in U_k\}$ is direct, written $U_1 \oplus \cdots \oplus U_m$, if $U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\}$ for $k = 2, \ldots, m$. In other words, $U_1 + U_2$ is direct, then $(U_1 + U_2) + U_3$ is direct as a sum of two subspaces, and so on up to U_m. Subspaces are never disjoint, since all contain 0; $U \cap W = \{0\}$ is the closest they can come.

Example: a direct sum in $\mathbb{R}^2$. Take the lines $U = \{(t, t)\}$ and $W = \{(s, -s)\}$. Their only common point is 0, so the sum is direct. Every vector splits into a piece in U and a piece in W in exactly one way: solving $(4, 2) = (t, t) + (s, -s)$ gives $t + s = 4$, $t - s = 2$, so $t = 3$, $s = 1$ and $(4, 2) = (3, 3) + (1, -1)$.



$(4, 2) = (3, 3) + (1, -1)$, with $(3, 3) \in U$ and $(1, -1) \in W$. The red arrow from $(3, 3)$ to $(4, 2)$ is the vector $(1, -1)$ drawn from a different starting point. Since $U \cap W = \{0\}$, no other split exists.

Theorem (Prop. 7 / LADR 1.41, 1.45)

For subspaces $U_1, \ldots, U_m$ of V the following are equivalent:

(i) the sum $U_1 + \cdots + U_m$ is direct;

(ii) every $v \in U_1 + \cdots + U_m$ is $u_1 + \cdots + u_m$ with $u_k \in U_k$ in exactly one way;

(iii) the only way to write $0 = u_1 + \cdots + u_m$ with $u_k \in U_k$ is with every $u_k = 0$.

The idea

Take two subspaces first. If $u + w = u' + w'$, then $u - u' = w' - w$. The left side is in U, the right side in W, so this vector is in $U \cap W$. If $U \cap W = \{0\}$ it is 0, and $u = u'$, $w = w'$. So "intersection 0" is exactly what makes the pieces of a vector unique. Conversely a nonzero $x \in U \cap W$ gives two ways to write 0: $0 + 0$ and $x + (-x)$.

For m subspaces do the same one subspace at a time, starting from the last: move everything except the U_m-parts to one side to get a vector in $U_m \cap (U_1 + \cdots + U_{m-1})$.

Proof. 1. **Assume (i): the sum is direct.** We show (ii), by induction on m; for $m = 1$ the equation $u_1 = \tilde{u}_1$ is the claim. A decomposition exists by the definition of the sum. Suppose

$$v = u_1 + \cdots + u_m = \tilde{u}_1 + \cdots + \tilde{u}_m.$$

Subtracting and regrouping,

$$u_m - \tilde{u}_m = (\tilde{u}_1 - u_1) + \cdots + (\tilde{u}_{m-1} - u_{m-1}).$$

The left side is in U_m and the right side in $U_1 + \cdots + U_{m-1}$, so both are in $U_m \cap (U_1 + \cdots + U_{m-1}) = \{0\}$. So $u_m = \tilde{u}_m$.

2. Now $u_1 + \cdots + u_{m-1} = \tilde{u}_1 + \cdots + \tilde{u}_{m-1}$ are two decompositions of one vector of $U_1 + \cdots + U_{m-1}$, and that sum is direct too (its conditions are those of Def. 6 for $k \leq m-1$). The induction hypothesis applied to this shorter direct sum gives $u_k = \tilde{u}_k$ for $k \leq m-1$.
3. **Assume (ii).** We show (iii): $0 = 0 + \cdots + 0$ is one way to write 0; by (ii) it is the only one.
4. **Assume (iii).** We show (i) by proving the contrapositive. If the sum is not direct, some $U_k \cap (U_1 + \cdots + U_{k-1})$ contains $u_k \neq 0$, say $u_k = u_1 + \cdots + u_{k-1}$ with $u_i \in U_i$. Then $u_1 + \cdots + u_{k-1} + (-u_k) + 0 + \cdots + 0 = 0$ with $-u_k \neq 0$, contradicting (iii).

□

Pairwise intersections are not enough. For $m \geq 3$, pairwise intersections being $\{0\}$ is *not* enough: the three lines in the figure (take $U_1 = \{(t,t)\}$, U_2 the x-axis, U_3 the y-axis) meet pairwise in $\{0\}$, but $(1,1) = (1,0) + (0,1)$ lies in $U_1 \cap (U_2 + U_3)$, and $0 = (1,1) - (1,0) - (0,1)$ is a nontrivial way to write 0.

Shortcuts

- **Subspace test in one line**: check $0 \in U$, then take arbitrary $v, w \in U$, $\lambda \in \mathbb{F}$ and check $\lambda v + w \in U$. Reason: Def. 5. Instance: $U = \{(x_1, x_2) : x_1 + 2x_2 = 0\}$ contains $(0,0)$, and if $v_1 + 2v_2 = 0$ and $w_1 + 2w_2 = 0$ then $(\lambda v_1 + w_1) + 2(\lambda v_2 + w_2) = \lambda \cdot 0 + 0 = 0$.
- **Sets defined by equations**: if every equation has the form $a_1 x_1 + \cdots + a_n x_n = 0$, the set is a subspace (the computation of the previous instance works for each equation). If some equation has a nonzero right side, 0 fails it, so the set is not a subspace. Equations with products or powers usually fail, but check: $x_1 x_2 = 0$ is the union of the two axes, not a subspace, while $x_1^2 = 0$ is the line $x_1 = 0$, a subspace.
- **Fastest disproof**: check 0 first. Reason: every subspace contains 0. Instance: $\{(x, y) : y = 2x + 1\}$ misses $(0,0)$; nothing else needs checking.
- **Direct sum of two subspaces**: check only $U \cap W = \{0\}$ (Def. 6). Instance: $U = \{(t,t)\}$, $W = \{(s,-s)\}$: $(t,t) = (s,-s)$ forces $t = s$ and $t = -s$, so $t = s = 0$.
- **Direct sum of three or more**: check that $0 = u_1 + \cdots + u_m$ with $u_k \in U_k$ forces every $u_k = 0$ (Prop. 7(iii)). Pairwise intersections are not enough (Traps). Instance: for the three axes in $\mathbb{R}^3$, the equation below forces $a = b = c = 0$, so the sum is direct:

$$0 = (a,0,0) + (0,b,0) + (0,0,c) = (a,b,c).$$

- **Over $\mathbb{F}_2$, closure under addition is enough**. Reason: the only scalars are 0 and 1, and $0 \cdot v = 0$, $1 \cdot v = v$. Condition: $\mathbb{F} = \mathbb{F}_2$ and $0 \in U$. Instance: $\{(0,0), (1,1)\} \subseteq \mathbb{F}_2^2$ is a subspace, since $(1,1) + (1,1) = (0,0)$.

Traps

- The union of two subspaces is usually not a subspace: the two axes in $\mathbb{R}^2$ contain $(1,0)$ and $(0,1)$ but not $(1,1)$.
- Closure under addition alone is not enough: the quadrant $\{(x,y) : x, y \geq 0\}$ is closed under addition, but $(-1)(1,0)$ leaves it. Closure under scalar multiplication alone is not enough either: the union of the two axes.
- $U_1 \cap (U_2 + U_3)$ need not equal $(U_1 \cap U_2) + (U_1 \cap U_3)$: for the three lines in the figure, the left side is U_1 and the right side is $\{0\}$.
- Pairwise $U_i \cap U_j = \{0\}$ does not make $U_1 + U_2 + U_3$ direct: for the same three lines, $0 = (1,1) - (1,0) - (0,1)$ with nonzero terms.
- Whether a set is a subspace depends on the field: $\mathbb{R}^2$ is closed under real multiples but not under multiplication by i (HW1 P6).

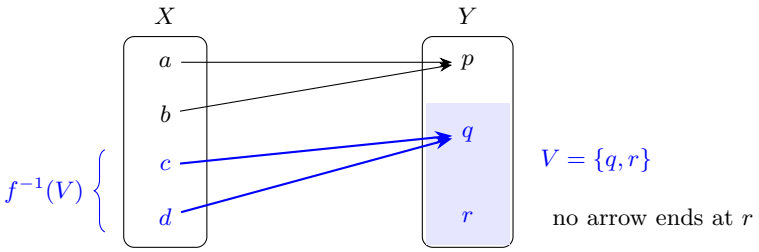
§2 Problem set

HW1 P1. Let $f\colon X \to Y$, $U \subseteq X$, $V \subseteq Y$.

(a) Prove $f(f^{-1}(V)) \subseteq V$.

(b) Prove $U \subseteq f^{-1}(f(U))$.

(c) Give examples where each inclusion is strict. Determine exactly when equality holds in (a) and in (b).



$f^{-1}(V)$ collects the starting points of the arrows that end in V: here c and d, both sent to q. No arrow ends at r, so $f(f^{-1}(V)) = \{q\}$ is smaller than V.

Hint: take an arbitrary element and unfold both definitions; for (c) think about surjective and injective.

HW1 P2 (LADR 1A.7). Show that $(-1 + \sqrt{3}\,i)/2$ is a cube root of 1.

Hint: square it first, then multiply once more; or use the shortcut with $x^2 + x + 1$.

HW1 P3. Let $a, b \in \mathbb{F}$. Prove that if $ab = 0$ then $a = 0$ or $b = 0$.

Hint: multiplicative inverse (field axiom (vii)).

HW1 P4 (LADR 1B.2). If $a \in \mathbb{F}$, $v \in V$ and $av = 0$, prove $a = 0$ or $v = 0$.

Hint: multiply by a^{-1} as in P3, then Prop. 4(ii).

HW1 P5 (LADR 1C.4). Let $b \in \mathbb{R}$. Show that the set of continuous $f\colon [0,1] \to \mathbb{R}$ with $\int_0^1 f = b$ is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Hint: the zero function for one direction, linearity of the integral for the other.

HW1 P6 (LADR 1C.5). Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Hint: which scalars are allowed in $\mathbb{C}^2$?

HW1 P7. Over $\mathbb{F}_2$, expand $(x+1)^8$ by computing $(x+1)^2$ and squaring repeatedly. Explain where the calculation differs from the one over $\mathbb{R}$.

Hint: $2 = 0$ in $\mathbb{F}_2$.

R1 (Midterm 1 review, Exercise 7). Prove that the intersection of subspaces is a subspace. What about the union?

Hint: check the two conditions of Def. 5 in each subspace (Prop. 5); for the union, try the two axes in $\mathbb{R}^2$.

R2 (Midterm 1 review, Exercise 14). Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$ there is a unique $\beta \in \mathbb{C}$ with $\alpha\beta = 1$.

Hint: existence from the conjugate formula for $1/(a+bi)$; uniqueness as in the uniqueness of inverses.

E1 (LADR 1C.12). Prove that the union of two subspaces of V is a subspace if and only if one of them contains the other.

Hint: if neither contains the other, pick $u \in U$ with $u \notin W$ and $w \in W$ with $w \notin U$, and ask which of U, W could contain $u + w$.

E2. In $\mathbb{R}^3$ let $U = \{(x, y, 0)\}$, $W = \{(0, y, z)\}$, $X = \{(t, t, t)\}$. Which of the sums $U + W$, $U + X$, $W + X$ are direct? Is $\mathbb{R}^3 = U \oplus X$?

Hint: for two subspaces, direct means the intersection is $\{0\}$ (Def. 6); for $U + X = \mathbb{R}^3$, solve $(a, b, c) = (x, y, 0) + (t, t, t)$.